

120 Days of Quantitative Finance

Month 4: Stochastic Calculus & Time Series Modeling

Week 14: SDEs and Black-Scholes

Day 93: Introduction to Stochastic Differential
Equations

1 Introduction

Stochastic Differential Equations (SDEs) are used to model systems that evolve over time under both deterministic trends and random shocks.

In financial markets, asset prices, interest rates, and volatility are commonly modeled using SDEs.

2 General Form of an SDE

A stochastic differential equation is written as:

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t$$

where:

- $\mu(X_t, t)$ is the **drift term** (deterministic component)
- $\sigma(X_t, t)$ is the **diffusion term** (random component)
- W_t is Brownian Motion

3 Interpretation

- dt represents deterministic evolution over time
- dW_t represents random shocks
- The system evolves as a combination of both

4 Connection to Discrete Models

The SDE can be approximated using:

$$X_{t+\Delta t} = X_t + \mu(X_t, t)\Delta t + \sigma(X_t, t)\sqrt{\Delta t}\epsilon_t$$

where:

$$\epsilon_t \sim \mathcal{N}(0, 1)$$

This connects continuous-time models to discrete simulations.

5 Examples

5.1 Arithmetic Brownian Motion

$$dX_t = \mu dt + \sigma dW_t$$

5.2 Geometric Brownian Motion

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

6 Expectation and Variance

For simple SDEs:

$$E[X_t] \quad \text{and} \quad Var(X_t)$$

can often be derived analytically.

7 Role of Ito Calculus

Since W_t is not differentiable, classical calculus does not apply.

Ito calculus provides the framework for:

- solving SDEs
- transforming processes
- deriving financial models

8 Why SDEs in Finance?

SDEs capture:

- randomness in price movements
- time evolution of markets
- uncertainty in future outcomes

They form the foundation of:

- Black-Scholes model
- stochastic volatility models
- interest rate models

9 Conclusion

Stochastic Differential Equations provide a powerful mathematical framework to describe dynamic systems under uncertainty.

They are central to modern quantitative finance.